

Curriculum Vitae

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Professional Objective

To apply my training in experimental physics, radio-frequency engineering, signal processing, and data-driven analysis to research and development roles in academia, national laboratories, and industry, with a focus on instrumentation, detector systems, and advanced measurement technologies.

Education

Ph.D. in Physics <i>2020–2026</i> University of Kansas, USA	3.964/4.0
M.S. in Electrical Engineering <i>2024–2025</i> University of Kansas, USA	3.778/4.0
M.Sc. in Physics <i>2018–2020</i> Aligarh Muslim University, India	9.170/10 (Gold Medalist)
B.Sc. Honors in Physics <i>2015–2018</i> Aligarh Muslim University, India	9.186/10 (Gold Medalist)
Higher Secondary (12th) <i>2014</i> Central Board of Secondary Education, India	9.120/10 (First Position)
Secondary (10th) <i>2012</i> Central Board of Secondary Education, India	10/10 (Certificate of Merit)

Research Experience

Selected Dissertations, Theses, and Projects

- **Physics PhD Dissertation** *2020–2026*
Title: “Toward a Five-Station Diffuse Ultra-High Energy Neutrino Search with the Askaryan Radio Array: ARA01 Analysis Framework”. Defended on May 8, 2026. Advised by Prof. D. Besson. Published on ProQuest ISBN: 9798252480190.
- **Electrical Engineering Master’s Project** *2024–2025*
Developed **AAFIYA**: **A**ntenna **A**nalysis in **F**requency-domain for **I**mpedance and **Y**ield **A**ssessment, a modular Python toolkit for rigorous, reproducible analysis and visualization of antenna S-parameters, impedance, and gain/beam patterns from both measurement and simulation data, with Prof. Jim Stiles at EECS Department, KU, (Report), (Slides).
- **Physics Master’s Thesis** *2019–2020*
Investigated light sterile neutrino oscillations with Prof. M.S. Athar at Aligarh Muslim University (AMU), producing review report “Light Sterile Neutrinos: Status and Prospects” (DOI:10.13140/RG.2.2.13777.66405).
- **Physics Undergraduate Project** *2017–2018*
“Quarks & Leptons”: Detailed study of Standard Model interactions with Prof. M.S. Athar at AMU [Handwritten Report].

Research Roles and Appointments

- **Analysis Lead, Radio Background Characterization with ARA** *Dec 2025–Present*
Leading the analysis of astrophysical, environmental, and anthropogenic radio backgrounds in ARA data with Prof. D. Besson at KU.
- **Primary Analyzer, Askaryan Radio Array (ARA) Station 1** *2021–2026*
Conducted data analysis of ARA Station 1 for ultra-high energy (UHE) neutrino detection with Prof. D. Besson at University of Kansas (KU).
- **IceCube Neutrino Observatory Monitoring** *2021–2026*
Served as KU representative for monitoring shifts, presenting weekly reports to the collaboration.
- **Machine Learning Project** *2025*
Developed a CNN model for “Automatic Surface Detection in Echograms using Deep Learning” as a part of Machine Learning class at EECS Department, KU, (Project Repository), (Report).
- **Visiting Summer Student Program** *May–Jul 2019*
Studied the role of θ_{13} mixing angle in three-flavor neutrino oscillations with Prof. S.K. Agarwalla at Institute of Physics, India, producing technical report “Importance of 1-3 Mixing Angle in Three-Flavor Neutrino Oscillations Paradigm” (DOI: 10.13140/RG.2.2.34421.12009).
- **Winter Internship Program** *Dec 2019–Jan 2020*
Researched “Weak Interaction Physics” at IIT-BHU with Dr. G. Abbas.

Field Experience

NSF’s Summit Station, Greenland Field Campaign *Apr–May 2025*

Participated in the Radio Neutrino Observatory – Greenland (RNO-G) calibration campaign, performing in-situ radio-frequency measurements and equipment deployment at Summit Station, Greenland.

Publications

First-Author Works

- **Seikh, M.F.H., Jarvis, R., & Stiles, J.** (2026). “AAFIYA: Antenna Analysis in Frequency-domain for Impedance and Yield Assessment”. arXiv:2601.13583 (Accepted in 2026 IEEE AP-S/URSI Symposium)
- **Seikh, M.F.H. & Besson, D.** (2026). “Polar Ice as a Laboratory for Physics”. *Page 27-37, Physics Bulletin 2026, Physics Department, AMU*
- **Seikh, M.F.H.** (2025). “A geometrical approach to neutrino oscillation parameters”. *AIP Advances 1 Oct 2025; 15 (10): 105204*. DOI:10.1063/5.0292233.
- **Seikh, M.F.H., Grachev, V., Kravchenko, I., Athar, M.S., & Besson, D.** 2025. “First Radio Neutrino & Cosmic Rays Astronomy Workshop in India”. *European Physics Journal-Special Topics (EPJ-ST)*. arXiv:2509.20704 DOI:10.1140/epjs/s11734-025-01955-8.
- **Seikh, M.F.H., Giri, P. & Besson, D. for the ARA Collaboration** 2025. “Techniques for Continuous Wave Identification and Filtering in the Askaryan Radio Array”. PoS(ICRC2025)1171. arXiv:2509.20735.
- **Seikh, M.F.H. & Besson, D. for the RNO-G Collaboration** 2025. “Observation of Radio Solar Flares in RNO-G”. PoS(ICRC2025)1365.
- **Seikh, M.F.H.** (2024). “Physics of radio antennas”. *European Physics Journal-Special Topics (EPJ-ST)*. arXiv:2411.07507 DOI: 10.1140/epjs/s11734-025-01500-7
- **Seikh, M.F.H. for the ARA Collaboration** (2024). “Askaryan Radio Array: searching for the highest energy neutrinos”. *European Physics Journal-Special Topics (EPJ-ST)*. arXiv:2411.01761 DOI: 10.1140/epjs/s11734-025-01708-7

- **Seikh, M.F.H. & Besson, D.** (2024). “IceCube Neutrino Observatory: A Cold Discovery Hub”. *Page 14-23, Physics Bulletin 2024, Physics Department, AMU*
- **Seikh, M.F.H. for the ARA Collaboration** (2023). “Calibration and Physics with ARA Station 1”. *PoS(ICRC2023)1163*. arXiv:2308.07292
- **Seikh, M.F.H.** (2020). “Neutrinos and the 40 Future Routes”. *Physics Bulletin*, Physics Department, AMU. DOI: 10.13140/RG.2.2.27981.10729

Collaborative Works

Full publication (includes contributions to ARA, RNO-G, ARIANNA, IceCube, IceCube Gen2, PUEO, TAROGE-M and RET collaborations) record available on:

Inspire HEP ; Google Scholar ; ResearchGate ; ORCID  ; Scopus

Professional Service

Editorial

- **Book Editor** for Springer’s EPJ-ST journal: “Radio Detection of Ultra-high Energy Neutrino & Cosmic Ray” (15 articles) with M.S. Athar, D. Besson, V. Grachev, and I. Karvchenko (2024–2025)
- **Founder and Editor-in-Chief** of *The Magnetic Monopole* e-magazine for KU’s Society of Physics Students [View Issue] (Jun 2022)

Research Article Review

- **Peer Reviewer:** for World Scientific’s IJMP-A journal (1 article, 2 reviews) (2025)
- **Peer Reviewer:** for Springer’s EPJ-ST journal (5 articles, 11 reviews) (2024–2025)
- **Internal ARA Collaboration Reviewer:** Pawan Giri for ARA Collaboration, “A Comprehensive Diffuse Neutrino Search Using the Full Askaryan Radio Array”, 32nd International Symposium on Lepton Photon Interactions at High Energies (Nov 2025)

Conference Session Chair

- **General Cosmology:** for APS Global Physics Summit (Mar 19, 2026)
- **Topical Groups Session II:** for APS Global Physics Summit (Mar 17, 2026)

Other Reviews

- **REU Application Reviewer:** for KU Physics and Astronomy (10 Applications) (2025)
- **SSGSA Application Reviewer:** for SSGSA (Multiple Applications) (2021–2026)

Professional Leadership/Membership

- **American Physical Society (APS):** Graduate Student Member (2020–Present)
- **National Society of Black Physicists:** Student Member (2021–Present)
- **Physics & Astronomy Graduate Student Organization (GSO), KU:**
 - ✓ Graduate Students Representative - Member at Large (2025–Present)
 - ✓ Graduate Students Representative - Liaison (2024–2025)
 - ✓ GSO Executive Officer (2023–2024)
- **Sir Syed Global Scholar Award (SSGSA):**
 - ✓ Peer Mentor (2023–Present)

- ✓ Team Lead, Promotions & Outreach (2023–2025)
- ✓ Member, Mentorship & Applications Team (2021–2023)
- ✓ AMU Campus Team Member (2019–2020)
- **Society of Physics Students, KU Chapter:** Graduate President (2021–2022)
- **Sigma Pi Sigma (National Physics Honor Society):** Lifetime Member
- **AMU Society of Physics:** Member (2018–Present)

Technical Proficiencies

- **Programming Languages:** C/C++, Python, FORTRAN
- **Physics Software:** CERN ROOT, AraRoot, Wolfram Mathematica
- **Simulation Packages:** AraSim, NuRadioMC, HFSS
- **Operating Systems:** Linux (Ubuntu), Windows, macOS
- **Document Preparation:** L^AT_EX, Microsoft Office Suite

Honors & Awards

Academic Honors

- **University Gold Medals:**
 - ✓ M.Sc. Physics (Top among 50 students), AMU (2020)
 - ✓ B.Sc. Honors Physics (Top among 120 students), AMU (2018)
 - ✓ Highest percentage in B.Sc. (Hons) Faculty of Science, AMU (2018)
- **Abdul Aziz Medal** for B.Sc. Physics, AMU (2018)
- **Undergraduate Topper Scholarship**, UGC India (2020)
- **University Merit Financial Award**, AMU (2019–2020)
- **First Position** in Higher Secondary (12th) among 40 students (2014)
- **Certificate of Merit** for perfect 10/10 CGPA in Secondary (10th) (2012)

Fellowships & Grants

- **Ralston Dissertation Fellowship**, KU Physics (2025–2026)
- **McKeithan Summer Fellowship**, KU Physics (2025)
- **Graduate Student Award for Distinguished Service**, KU (2024–2025)
- **Wallace S. Strobel Scholarship**, KU EECS (2024–2025)
- **Ralston Summer Fellowship**, KU Physics (2024)
- **Doctoral Student Research Fund**, KU (2024)
- **Distinguished Students Award**, APS FIP (2021–2022)
- **Carl & Catherine Chaffee Scholarship**, KU Physics (2020–2021)

Conference Travel Awards

- **Junior Participant Support**, Neutrino (2026)
- **Graduate Student Travel Fund**, KU Physics (2023, 2025)
- **DAP Travel Grant**, APS (2024)
- **DPF Travel Award**, APS (2022, 2024)
- **CLAS Graduate Scholarly Development Fund**, KU (2022)

Other Distinctions

- **Washington DC Scholarship**, AMU Alumni Association (2015–2019)
- **Sir Syed Global Scholar Award** for prospective graduate students (2018)

Invited Lectures, Seminars, & Colloquia

- **Searching for the Universe’s Highest-Energy Neutrinos with Radio Detectors**
Particle Physics Seminar, Carnegie Mellon University (Sep 9, 2026)
- **Radio Engineering for Particle Astrophysics Experiments**
HEP Seminar, University of Nebraska-Lincoln (Mar 26, 2025)
- **Radio Engineering for Particle Astrophysics Experiments**
Astroparticle Seminar, Ohio State University (Jan 22, 2025)
- **Radio Engineering & Antenna Physics**
Lecture Series, IEI Summer School 2024, University of Alaska, Anchorage (Jul 2–5, 2024)
- **UHE Neutrino & Cosmic Ray Detectors**
ICEHAP Seminar, Chiba University, Japan (Aug 9, 2023)
- **Antenna Physics & Measurements**
Lecture Series, IEI Summer School 2023, South Dakota School of Mines (Jul 2–5, 2023)
- **UHE Neutrino & Cosmic Ray Detectors**
Physics Colloquium, AMU, India (Jan 12, 2023)
- **Antennas in UHE Neutrino Detectors**
Particle Physics Seminar, University of Delaware (Oct 4, 2022)

Outreach & Leadership

- Co-organized and hosted India’s inaugural workshop on **Radio Detection of UHE Neutrino and Cosmic Rays** at AMU (Apr 2024)
- Conducted outreach seminar on **Physics Research Opportunities** at AMU (2021)
- Participated in India’s first **Science Leadership Workshop** (2020)
- Volunteer for **One Person One Tree** plantation drive (2017)
- Participant in **Blood Donation Camp** (2017)

Conference Presentations

Oral Presentations

- **Pioneering an Array-Wide Search for Ultra-high Energy Neutrinos with ARA**
APS Global Physics Summit (Mar 18, 2026)
- **RNO-G in-ice Antennas**
RNO-G Collaboration Meeting, Uppsala University (Sep 9–14, 2024)
- **Radio Frequency Instrumentation**
Radio Neutrino Workshop, AMU, India (Apr 18–20, 2024)
- **Towards Five Station ARA Analysis**
APS April Meeting, Sacramento (Apr 3–6, 2024)
- **ARA Antenna Models**
ARA Collaboration Meeting, Ohio State University (Mar 5–8, 2024)
- **Current Status & Future of ARA**
APS Prairie Section Meeting, University of Missouri (Nov 30–Dec 2, 2023)

- **Introduction to Antenna Physics**
RET Collaboration Meeting, Vrije University Brussels (*Nov 8–11, 2023*)
- **Gain Calibration of RNO-G Antennas**
RNO-G Fall Meeting, Penn State University (*Sep 11–15, 2023*)
- **Galactic Noise in Radio Neutrino Experiments**
Astro Seminar, KU (*Oct 28, 2022*)
- **Towards ARA Station 1 Calibration**
APS April Meeting, New York City (*Apr 9–12, 2022*)
- **RNO-G Ice Calibration Planning**
RNO-G Collaboration Meeting, University of Chicago (*Apr 8, 2022*)

Poster Presentations

- **A Diffuse Search for UHE Cosmogenic Neutrinos with ARA Five Stations**
Neutrino 2026, UC Irvine (*Jun 21–26, 2026*)
- **Monitoring Snow Accumulation in ARA Using DAQ Temperature Data**
Neutrino 2026, UC Irvine (*Jun 21–26, 2026*)
- **A Geometrical Approach to Neutrino Oscillation Parameters**
APS Global Physics Summit (*Mar 17, 2026*)
- **CW Identification & Filtering Techniques in ARA**
39th ICRC, Geneva, Switzerland (*Jul 19, 2025*)
- **Observation of radio solar flares in RNO-G**
39th ICRC, Geneva, Switzerland (*Jul 15, 2025*)
- **Gain Measurements of LPDA Antennas**
APS April Meeting, Sacramento (*Apr 3–6, 2024*)
- **Overview of ARA Detector: Five Station Analysis**
ICTS Neutrino School, India (*Apr 22–May 3, 2024*)
- **Calibration of ARA Station 1**
38th ICRC, Nagoya University, Japan (*Jul 29, 2023*)

Professional Development

- **The AI Masterclass 3.0**, (Dhruv Rathee Academy, *May 2026*)
- **Responsible Conduct of Research**, (CITI Program, *2021, 2025*) - 100%
- **From the Big Bang to Dark Energy** (U Tokyo Coursera, *2020*) - 100%
- **Particle Physics: Introduction** (U Geneva Coursera, *2019*) - 100%
- **Senior Diploma in Painting** (Bangiya Sangeet Parishad) - Distinction

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Special Topics

Radio Detection of Ultra-high Energy Neutrino and Cosmic Ray

Mohammad Ful Hossain Seikh, Mohammad Sajjad Athar,
Dave Besson, Victor Grachev, and Ilya Kravchenko (Eds.)



Cosmic rays and neutrinos: exploring high-energy messengers through radio detection.
Courtesy of Mohammad Ful Hossain Seikh

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